Annotated Wireframes

Part 1 · Getting in

Screens 1–4. Opening the app, making an account and choosing a grade.

1. Start (Splash screen)

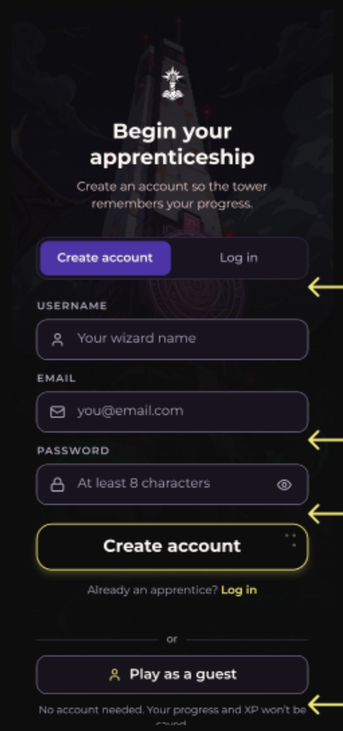


Logo and tagline set the magic-maths theme

Animated tower art plays behind

One big button starts the game

2. Sign up (Create account screen)



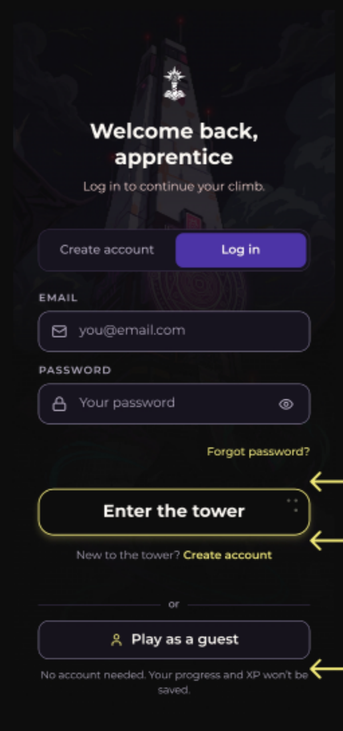
Tabs switch between sign up and log in

Short form: name, email, password

Eye icon shows or hides the password

Play as a guest skips making an account

3. Log in (Log in screen)

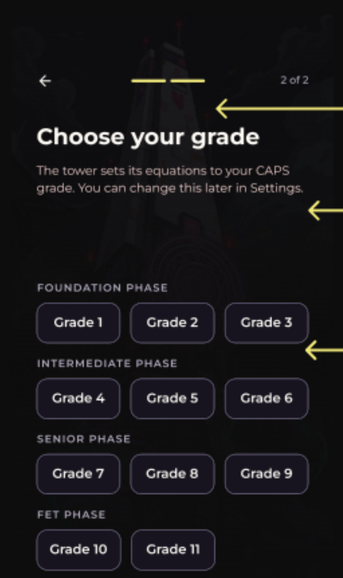


Forgot password sends a reset code by email

Big gold button logs you straight in

Guest play: no account, nothing saved

4. Onboarding (Choose grade screen)



Step bar shows setup progress

Explains the grade sets the maths level

Grades grouped by school phase, one tap each

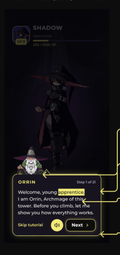
Button stays dim until a grade is picked

Annotated Wireframes

Part 2 · Tutorial

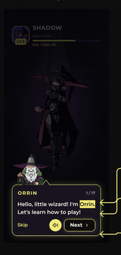
Screens 5–25. The full senior tutorial (21 steps). The junior tutorial (JS) teaches the same ideas in 17 shorter steps with fewer words. Each step lights up one part of the real screen.

5. Tutorial (Step 1 - Welcome screen)

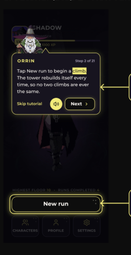


53. Tutorial (Welcome screen)

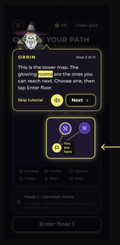
JUNIOR MODE



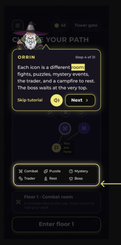
6. Tutorial (Step 2 - New run screen)



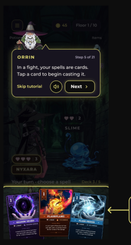
7. Tutorial (Step 3 - Tower map screen)



8. Tutorial (Step 4 - Room types screen)



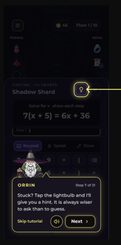
9. Tutorial (Step 5 - Spell cards screen)



10. Tutorial (Step 6 - Solve the sum screen)



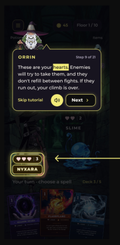
11. Tutorial (Step 7 - Hints screen)



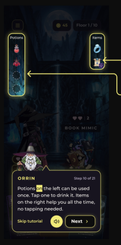
12. Tutorial (Step 8 - Who goes first screen)



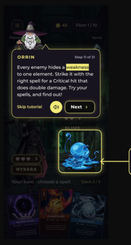
13. Tutorial (Step 9 - Hearts screen)



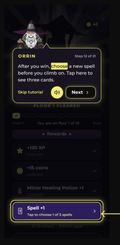
14. Tutorial (Step 10 - Potions and items screen)



15. Tutorial (Step 11 - Weaknesses screen)



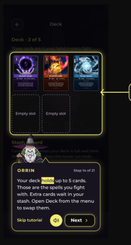
16. Tutorial (Step 12 - New spell reward screen)



17. Tutorial (Step 13 - Reading a card screen)



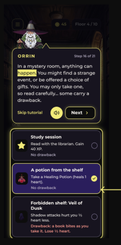
18. Tutorial (Step 14 - Deck screen)



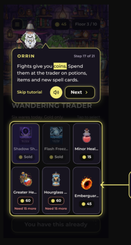
19. Tutorial (Step 15 - Puzzle rooms screen)



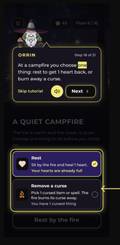
20. Tutorial (Step 16 - Mystery rooms screen)



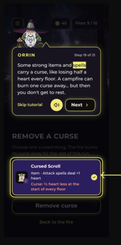
21. Tutorial (Step 17 - Trader screen)



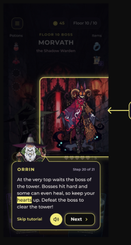
22. Tutorial (Step 18 - Campfire screen)



23. Tutorial (Step 19 - Curses screen)



24. Tutorial (Step 20 - Boss screen)



25. Tutorial (Step 21 - Ready to climb screen)

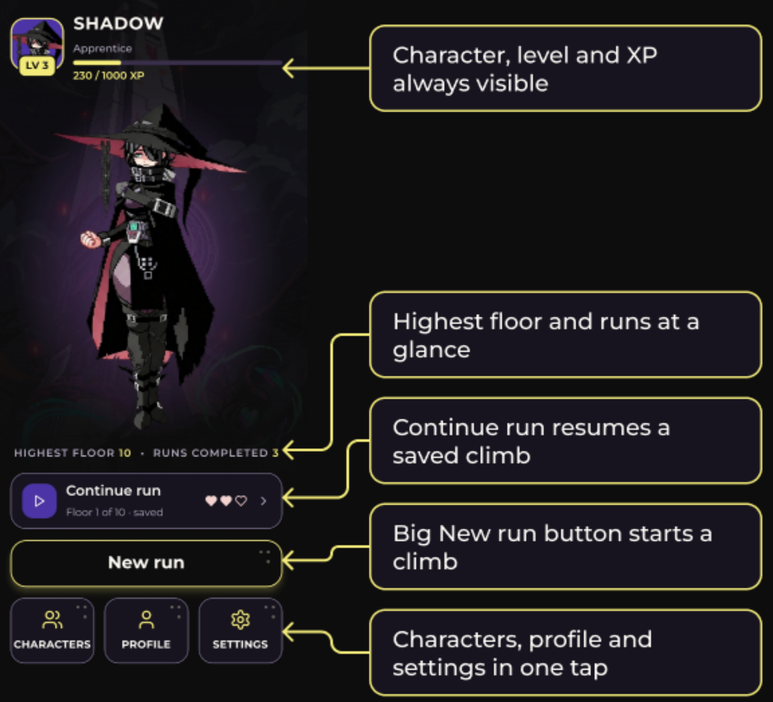


Annotated Wireframes

Part 3 · Home and settings

Screens 26–31. The home base between climbs: characters, profile, account and settings.

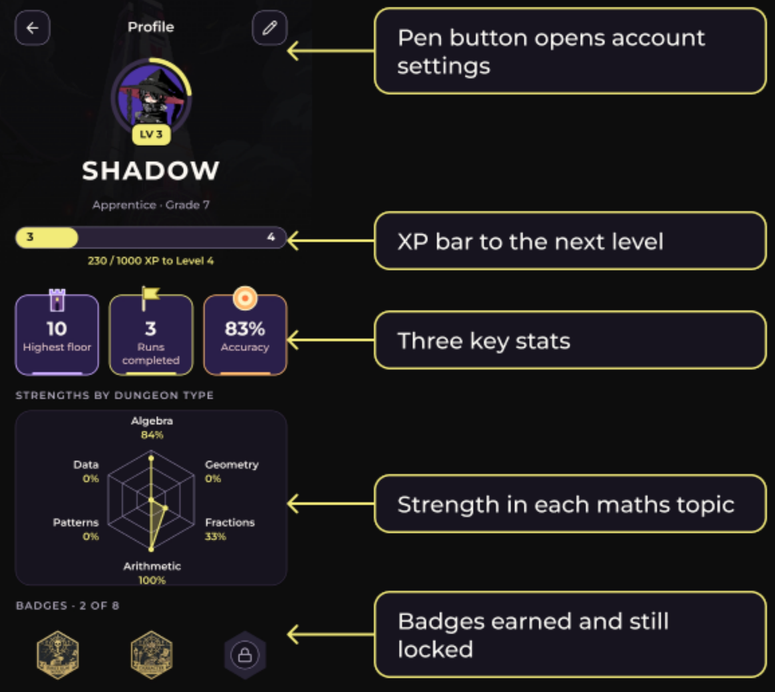
26. Home (Home screen)



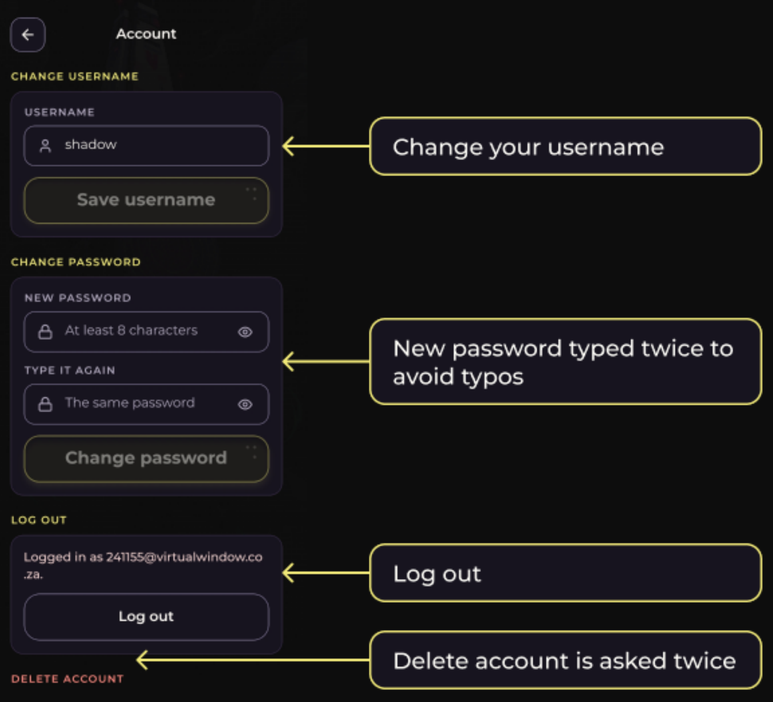
27. Characters (Character select screen)



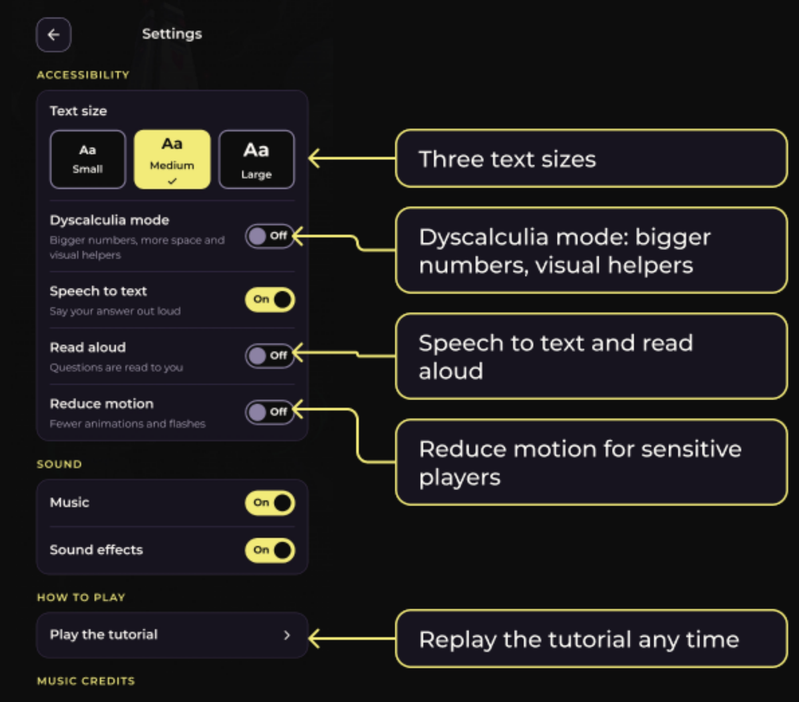
28. Profile (Profile screen)



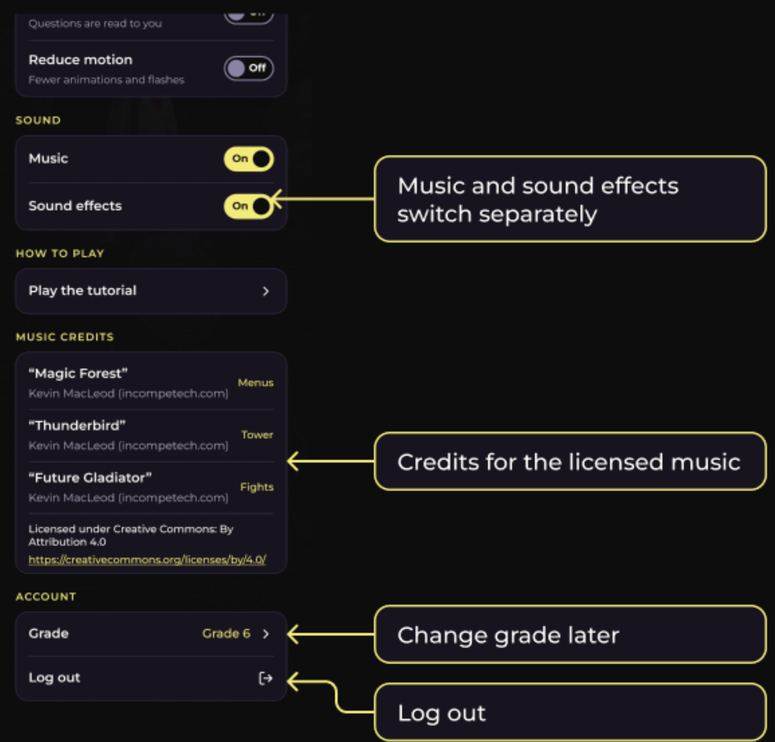
29. Account (Account settings screen)



30. Settings (Settings screen)



31. Settings (Sound, credits and account screen)

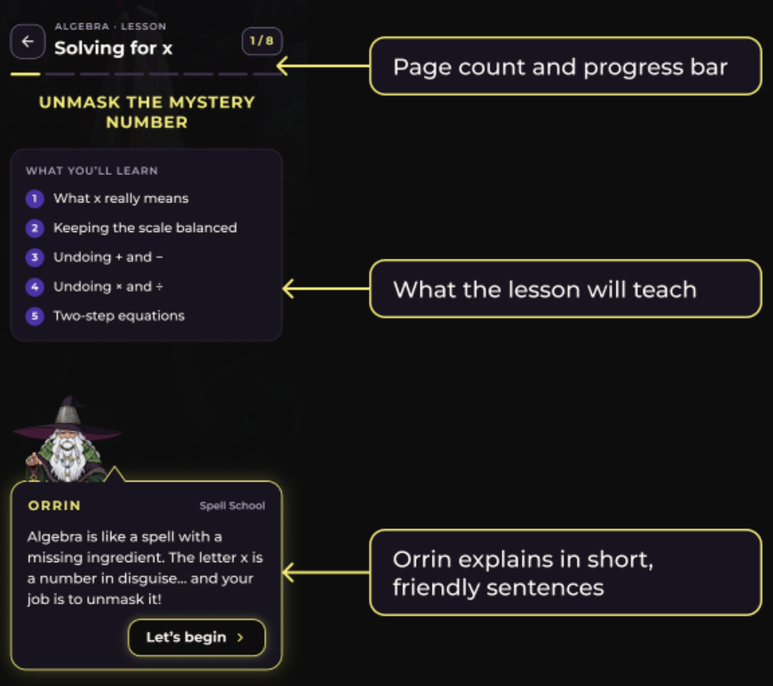


Annotated Wireframes

Part 4 · Spell School

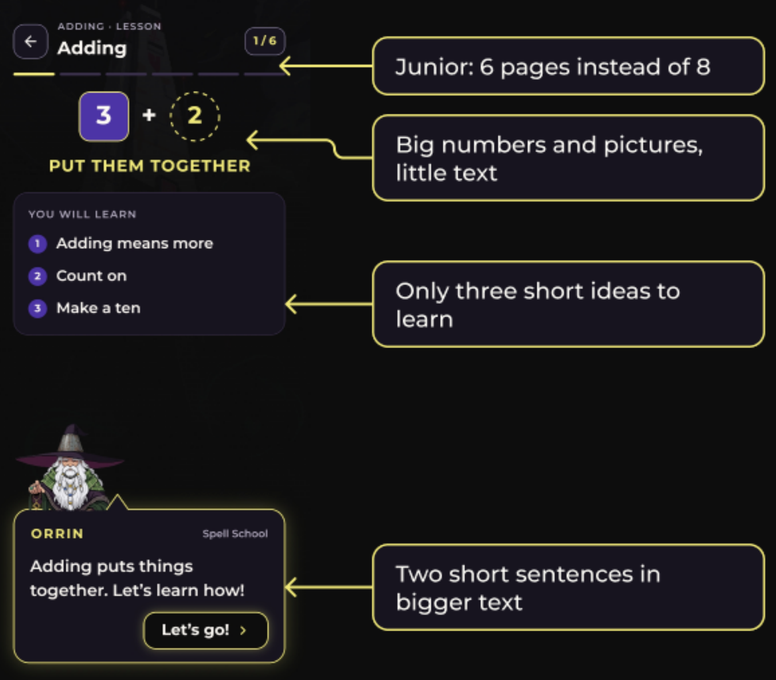
Screens 32–40. A short lesson for every maths topic: Orrin explains, shows worked examples, then the player tries one. Junior lessons (32J) use fewer pages, bigger pictures and less text.

32. Spell School (Lesson intro screen)

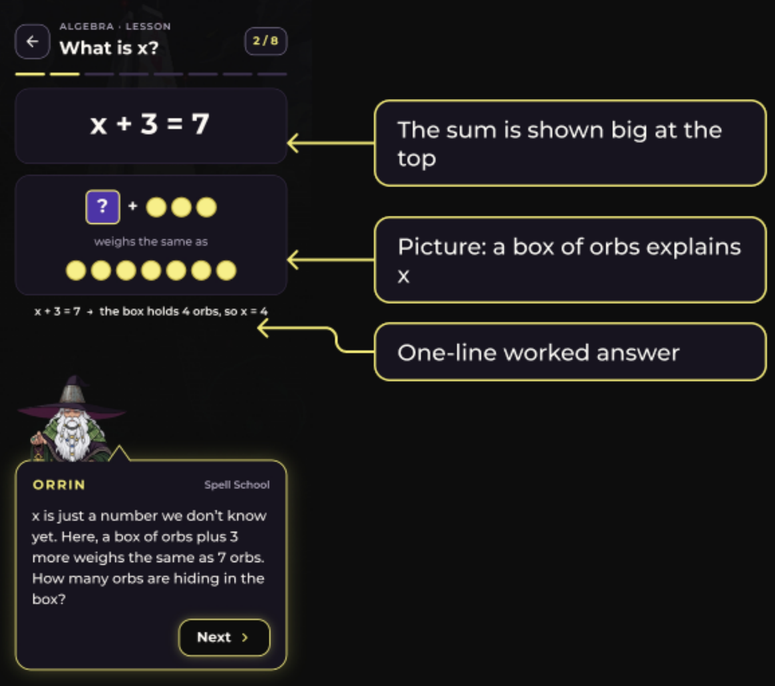


JUNIOR MODE

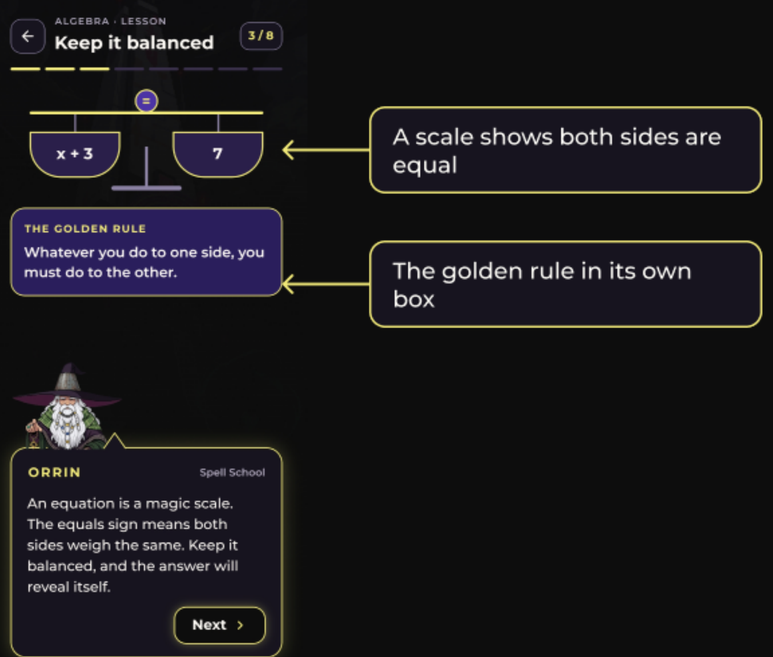
32J. Spell School (Lesson intro screen)



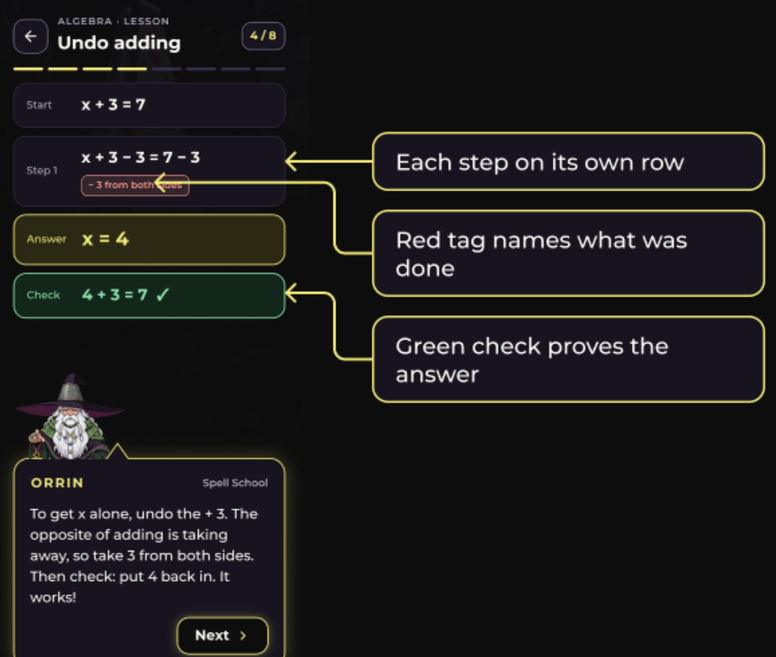
33. Spell School (What is x screen)



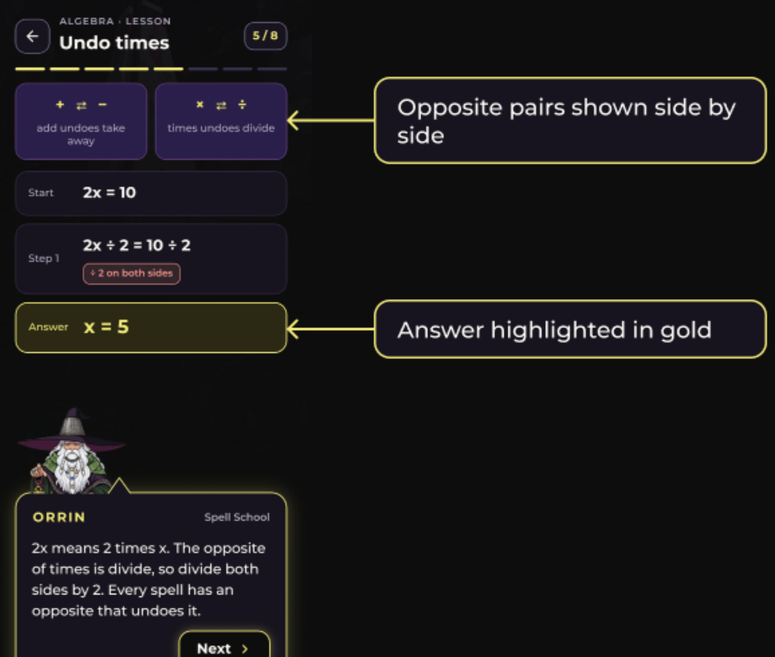
34. Spell School (Keep it balanced screen)



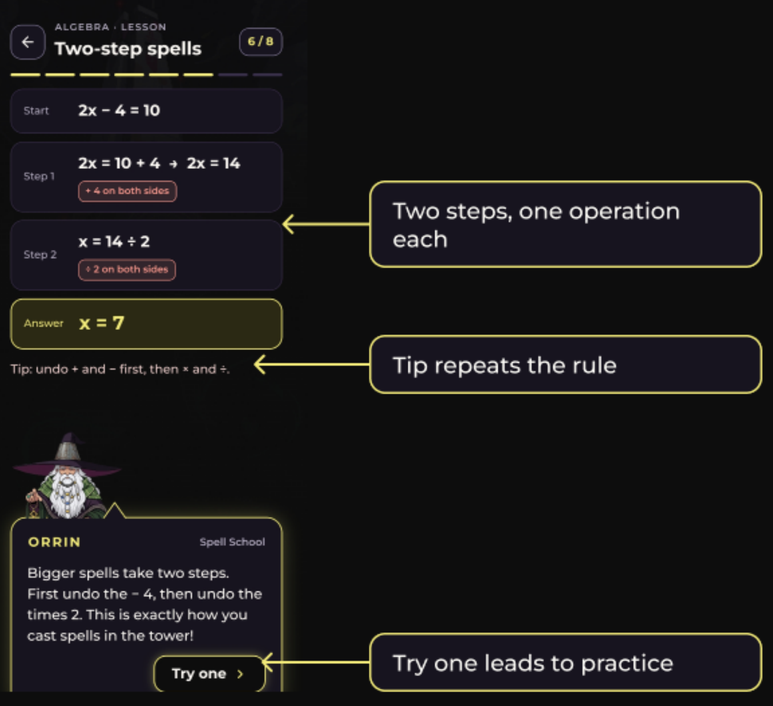
35. Spell School (Undo adding screen)



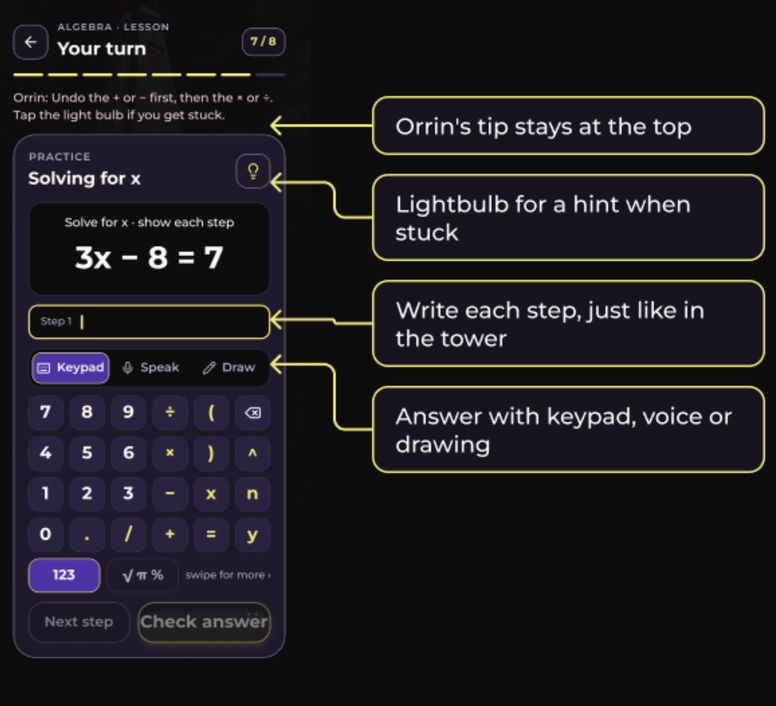
36. Spell School (Undo times screen)



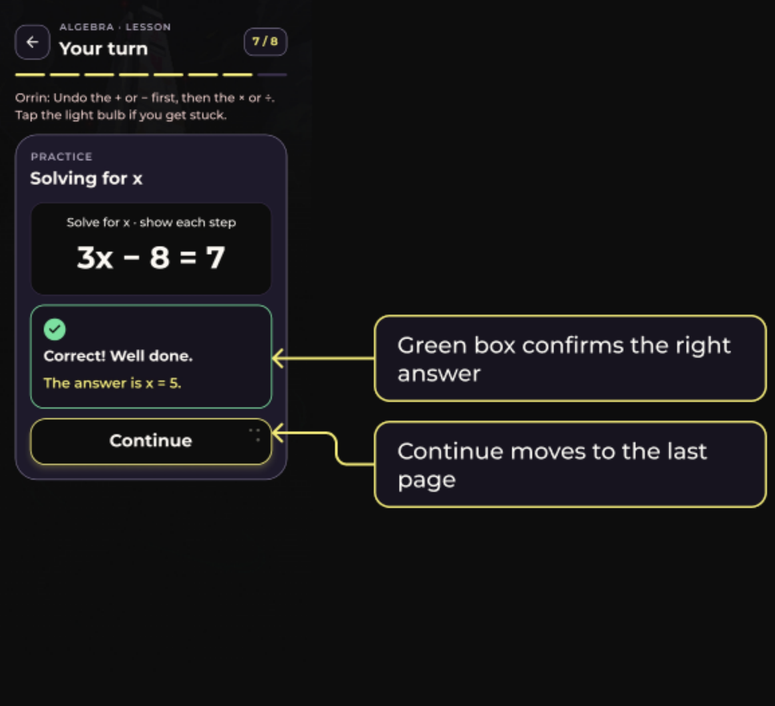
37. Spell School (Two-step spells screen)



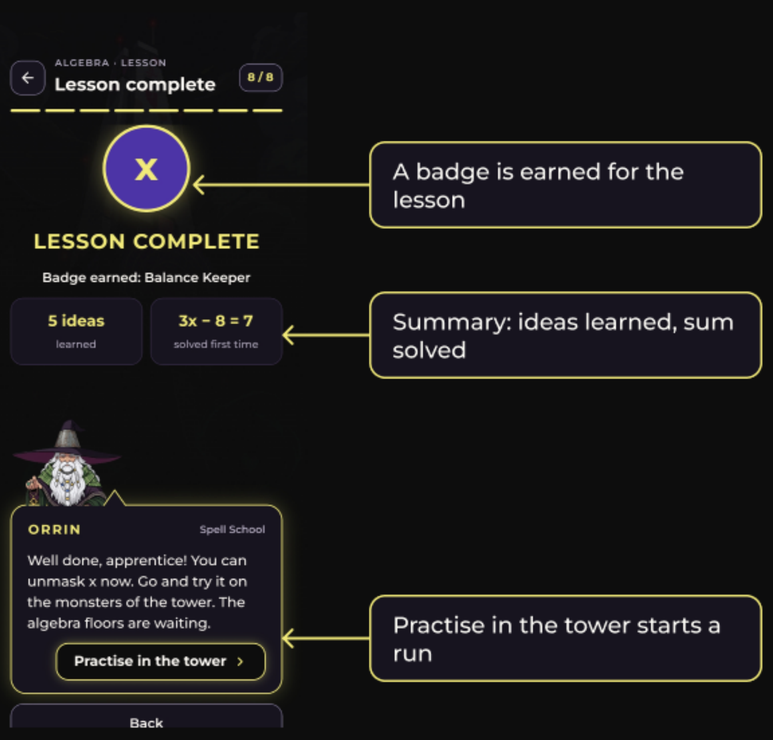
38. Spell School (Practice screen)



39. Spell School (Practice correct screen)



40. Spell School (Lesson complete screen)

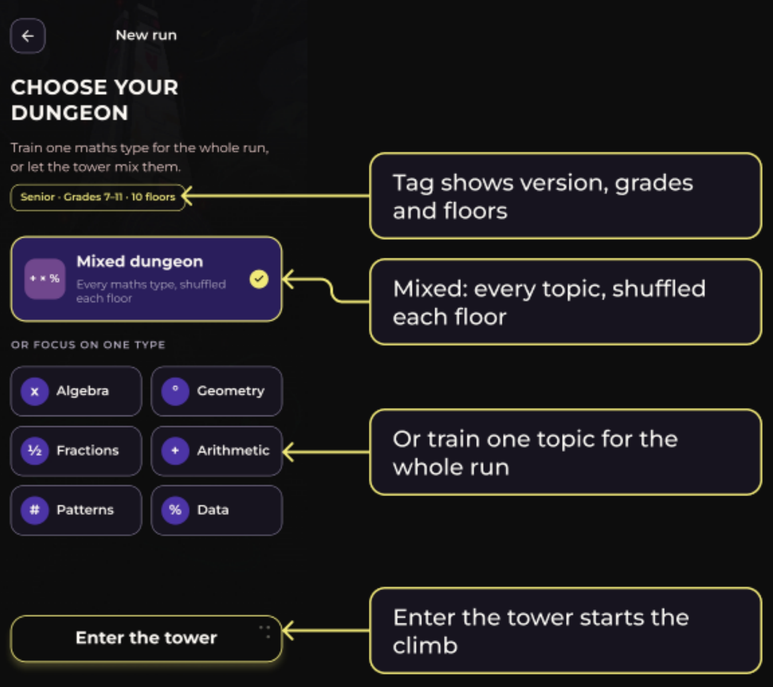


Annotated Wireframes

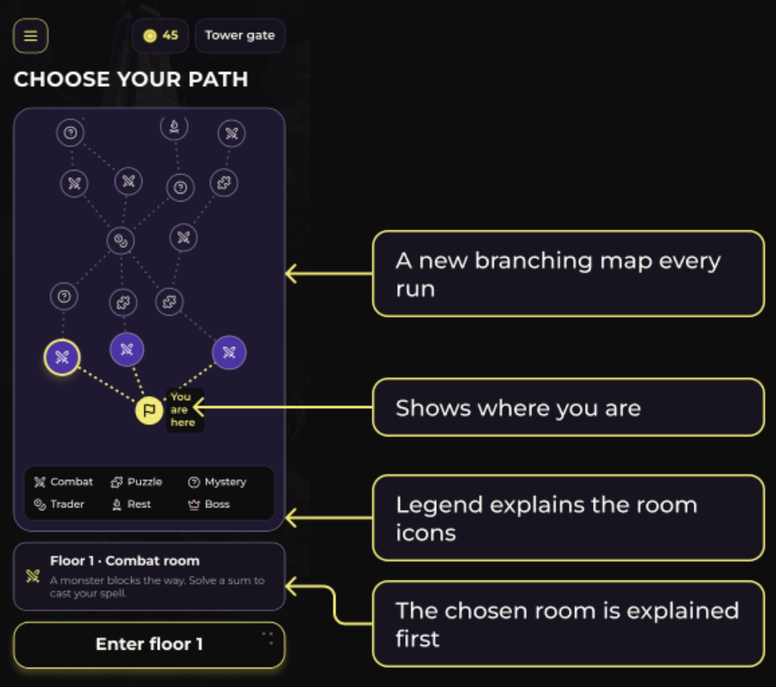
Part 5 · A fight: answering spells

Screens 41–49. Choosing a dungeon and map, then casting spells. Senior writes each step, intermediate types the final answer, junior taps one of three answers.

41. New run (Choose dungeon screen)

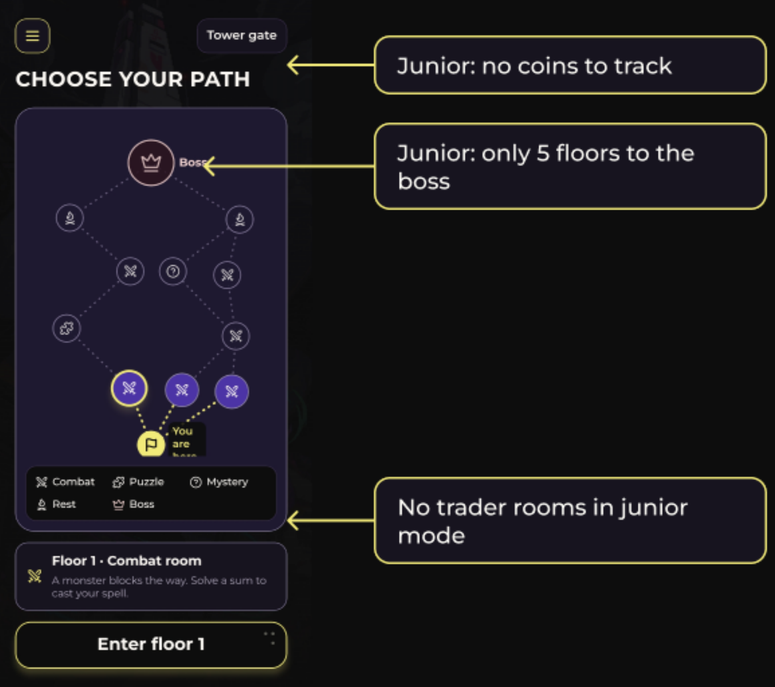


42. Tower map (Choose your path screen)

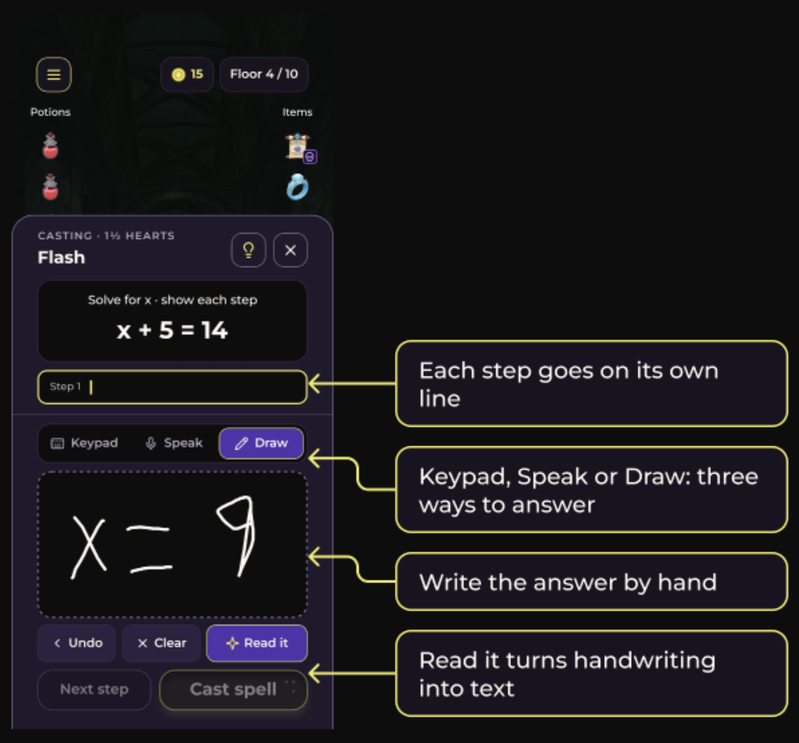


JUNIOR MODE

42J. Tower map (Choose your path screen)



43. Fight (Draw your answer screen)

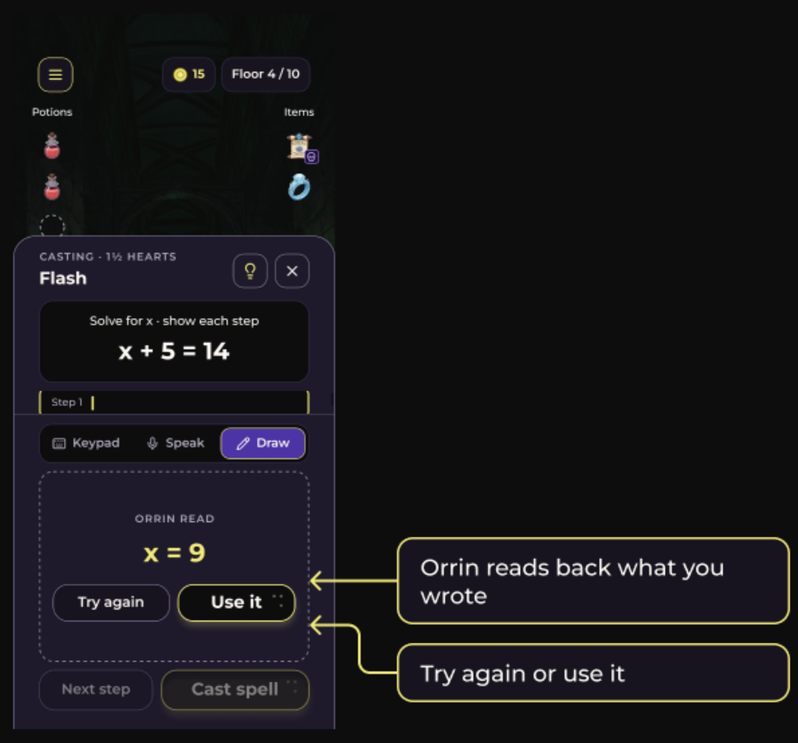


JUNIOR MODE

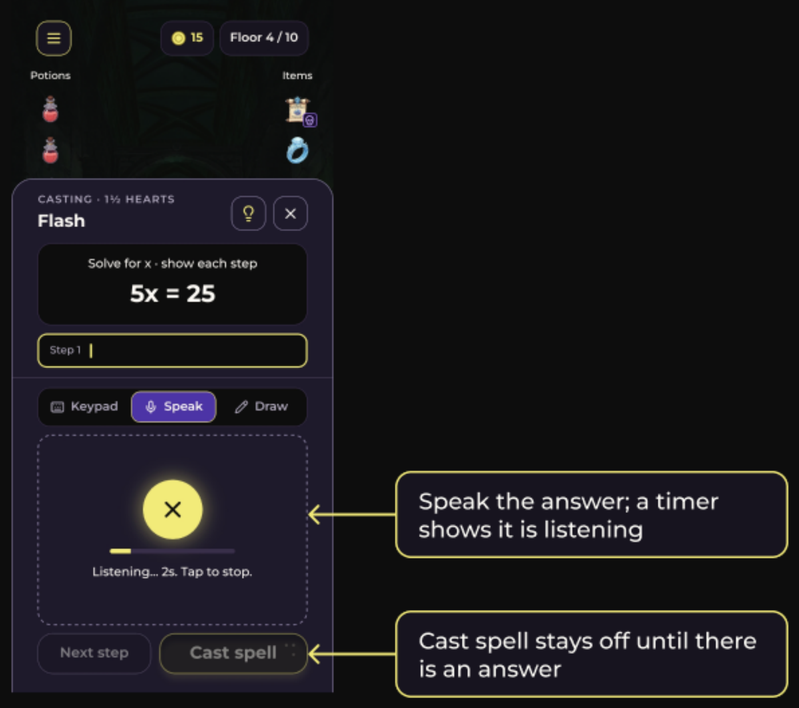
43J. Fight (Pick a spell screen)



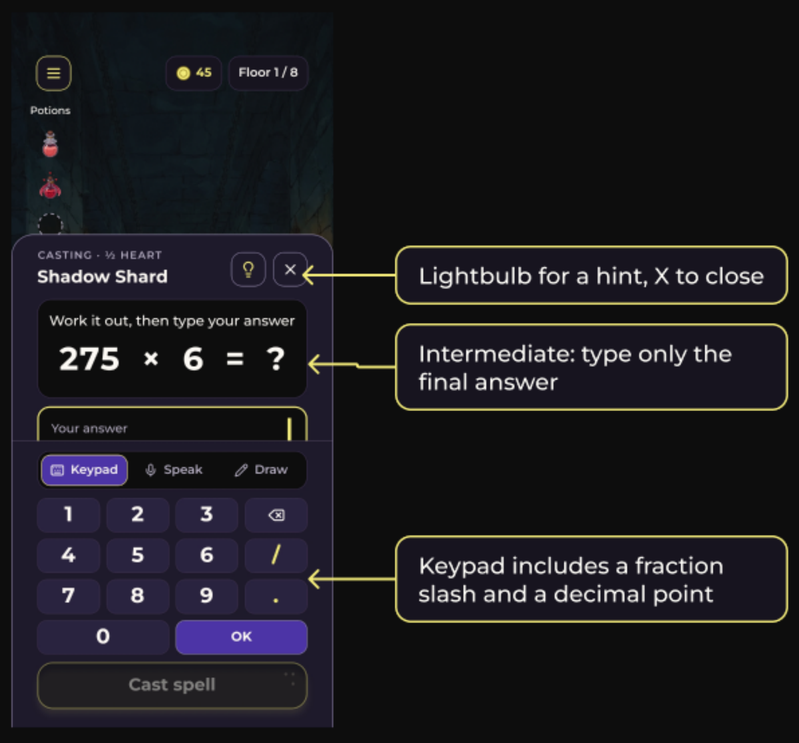
44. Fight (Handwriting read screen)



45. Fight (Speak your answer screen)

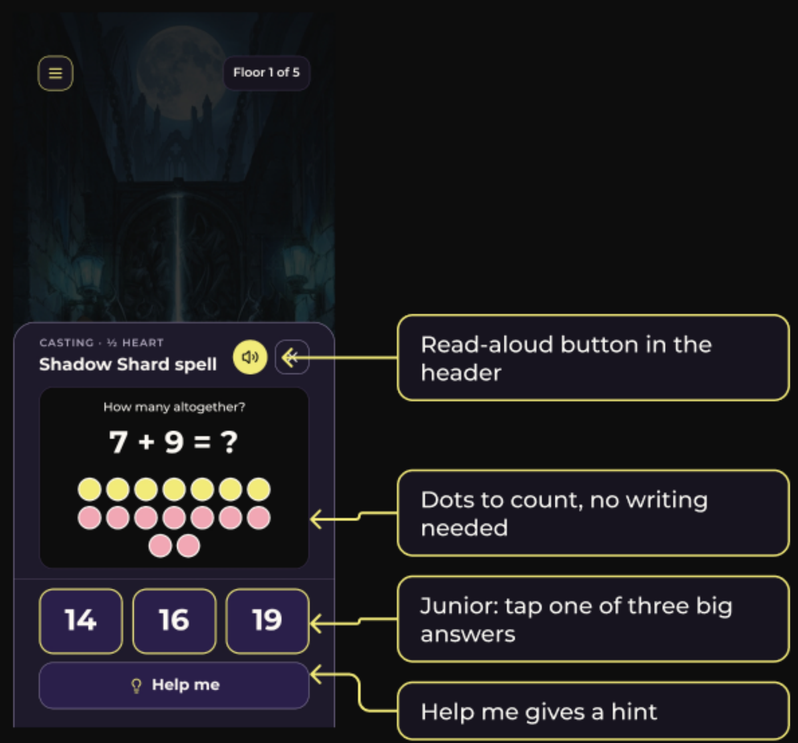


46. Fight (Type your answer screen)

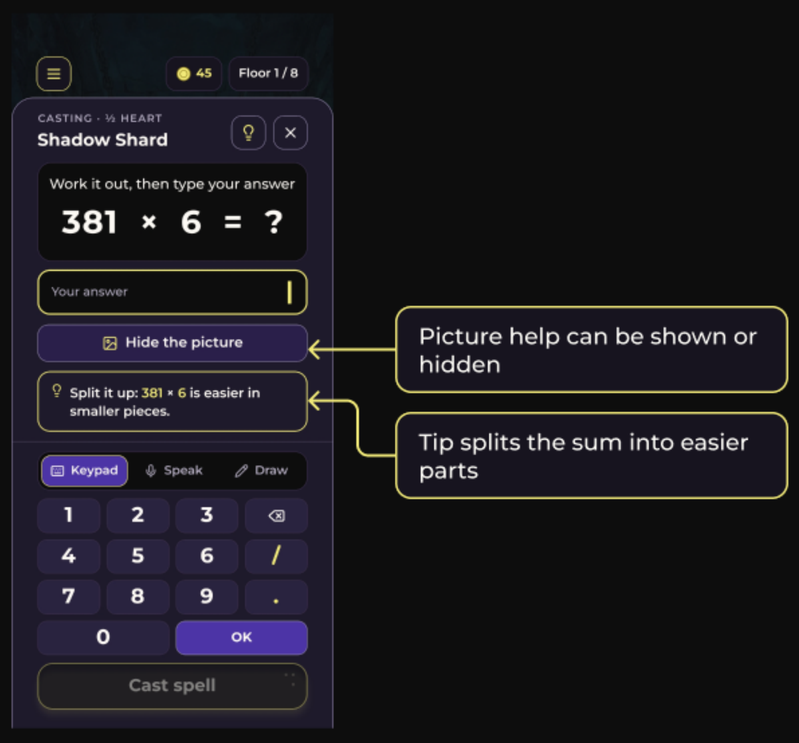


JUNIOR MODE

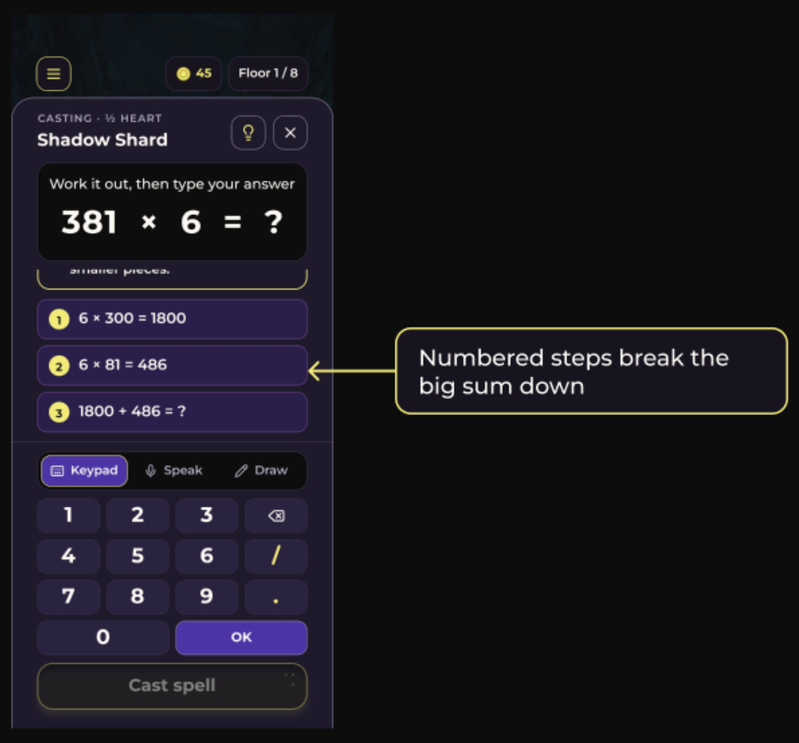
46J. Fight (Tap your answer screen)



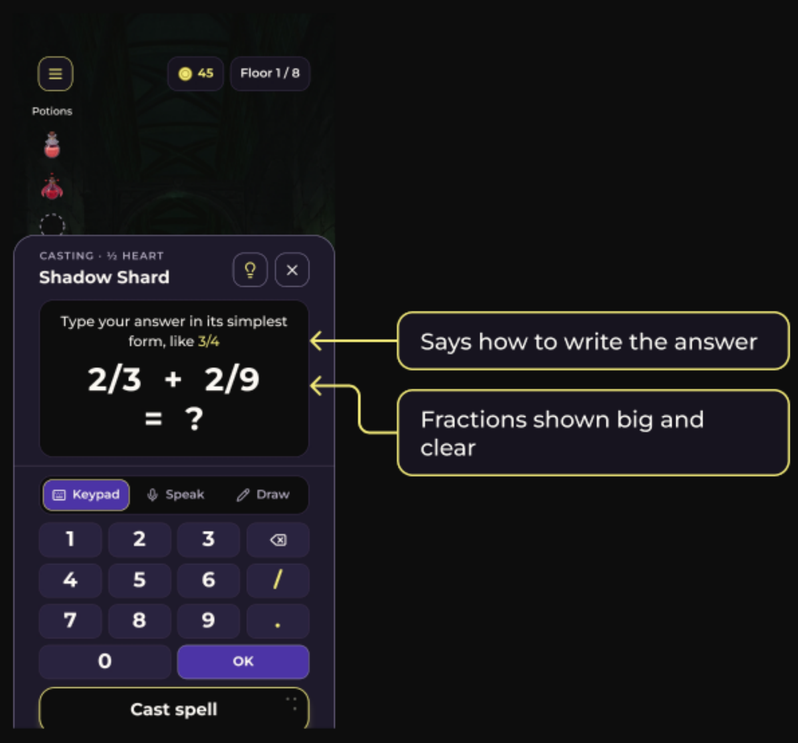
47. Fight (Show me a picture screen)



48. Fight (Picture steps screen)



49. Fight (Fraction answer screen)



Annotated Wireframes

Part 6 · Winning a fight

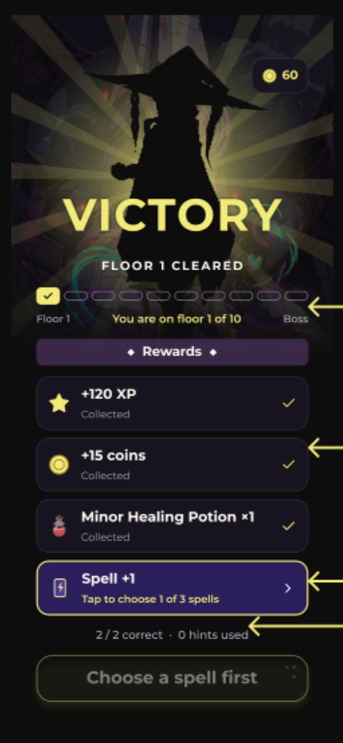
Screens 50–56. Feedback after a fight, rewards, badges, the run menu and the deck. Junior rewards (51J, 53J) use stars and simpler words.

50. Fight (Enemy defeated screen)



- A toast message confirms what happened
- The enemy's hearts are empty
- Your hearts after the fight

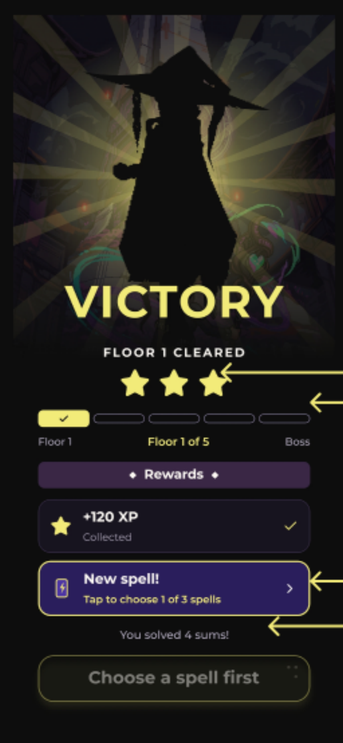
51. Rewards (Victory screen)



- Progress bar: how far up the tower
- Rewards ticked off as they are collected
- Pick a spell before moving on
- Correct answers and hints used

JUNIOR MODE

51J. Rewards (Victory screen)



- Junior: stars instead of numbers
- A shorter tower of 5 floors
- Simpler words: "New spell!"
- Friendly praise, no hint count

52. Badges (Badge unlocked screen)



- Badge artwork
- Badge name and how it was earned
- Pops up once, on a calm screen

53. Rewards (Choose a spell screen)



- Arrows flip through three offers
- Full card shows element, level and damage
- Dots show which card you are on
- Add to deck takes the card

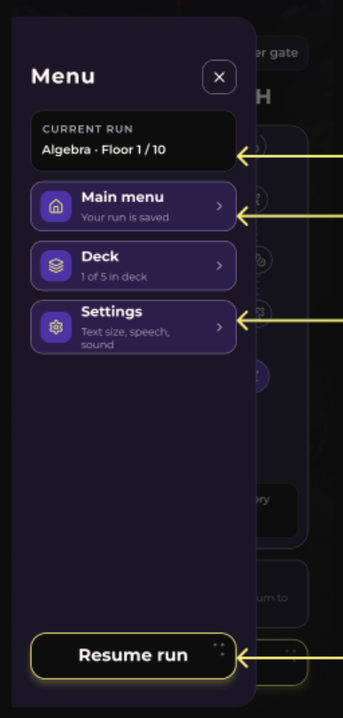
JUNIOR MODE

53J. Rewards (Choose a spell screen)



- Junior card: simple words, total damage
- Same layout as senior, less text

54. Run menu (Menu screen)



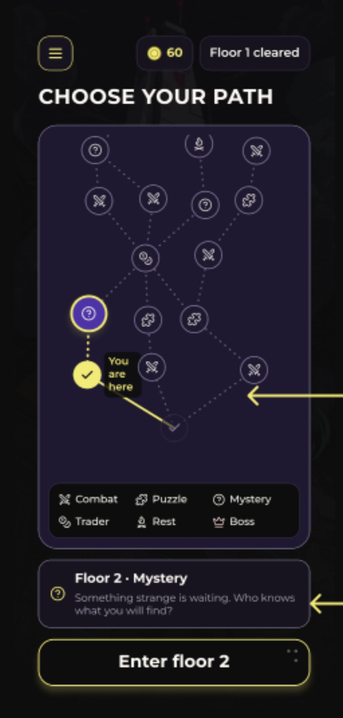
- Shows the current run
- Main menu: the run is saved
- Deck and settings open over the room
- Resume run closes the menu

55. Deck (Deck screen)



- Deck: up to 5 cards in every fight
- Empty slots are dashed
- The stash holds extra cards
- Tap a card to move it

56. Tower map (Next floor screen)



- The path behind you is ticked off
- The next room is explained first

Annotated Wireframes

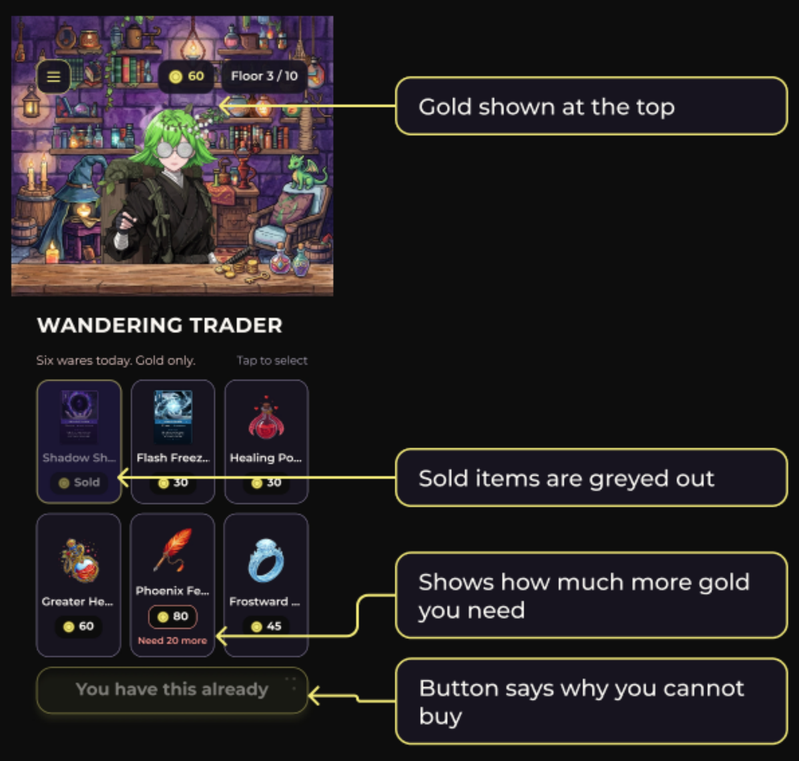
Part 7 · Other rooms

Screens 57–64. Mystery events, the trader, potions, items, the campfire and puzzle doors.

57. Mystery room (Wandering ghost screen)



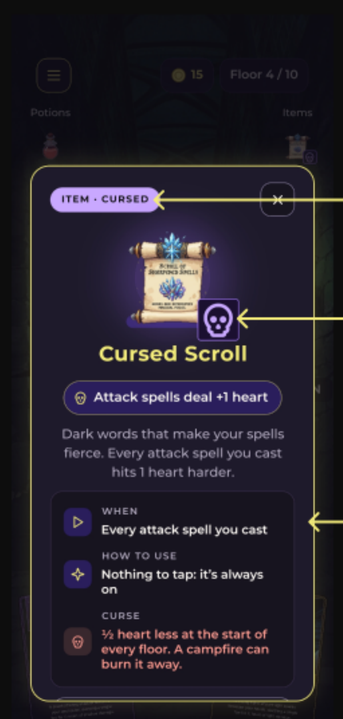
58. Trader room (Shop screen)



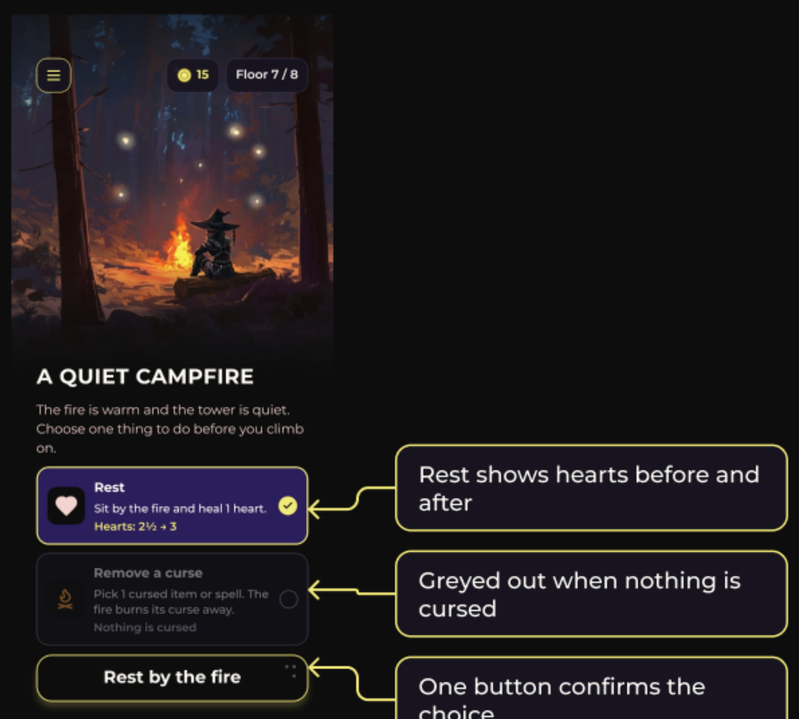
59. Items (Potion details screen)



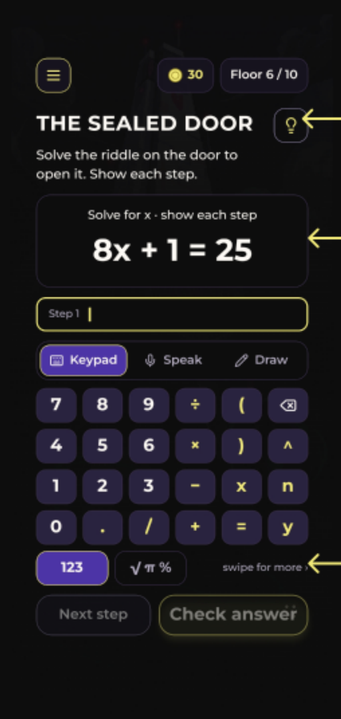
60. Items (Cursed item screen)



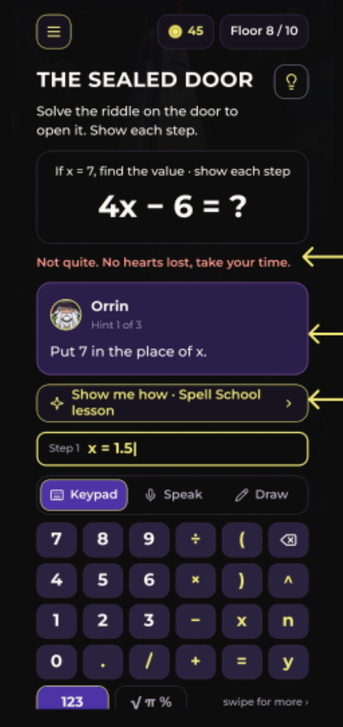
61. Campfire (Rest site screen)



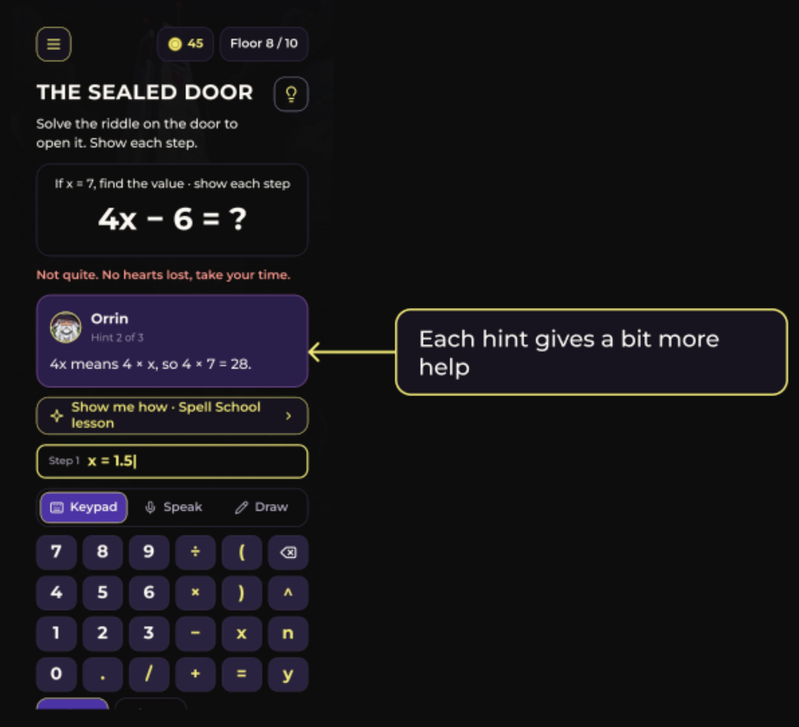
62. Puzzle room (Sealed door screen)



63. Puzzle room (Wrong answer hint screen)



64. Puzzle room (Second hint screen)

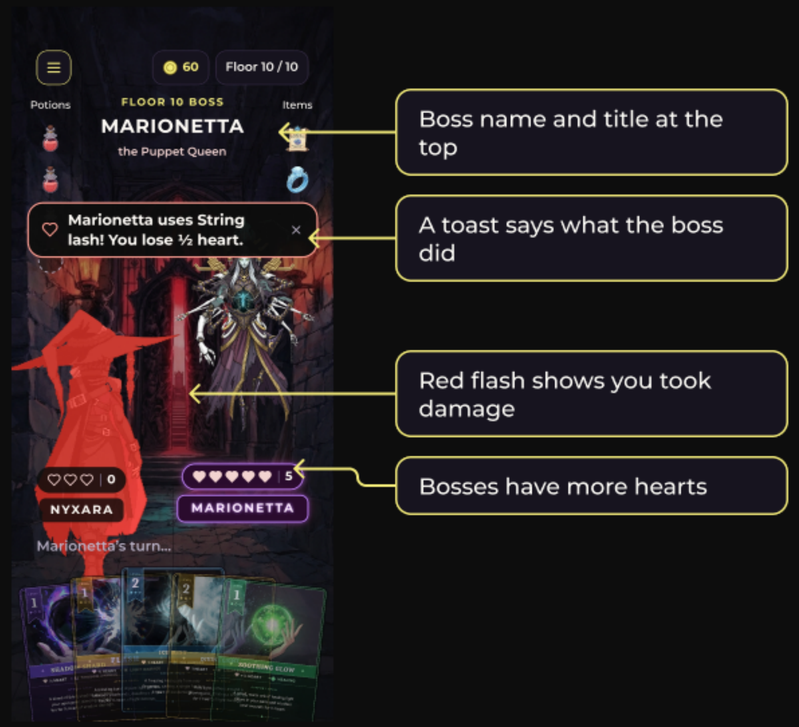


Annotated Wireframes

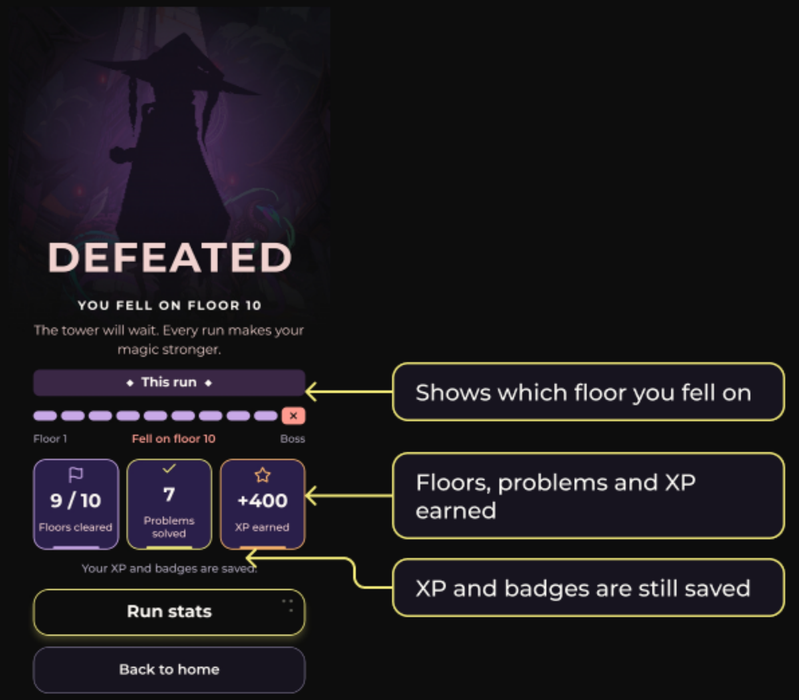
Part 8 · Boss and end of run

Screens 65–67. The boss fight, being defeated, and the run summary with advice.

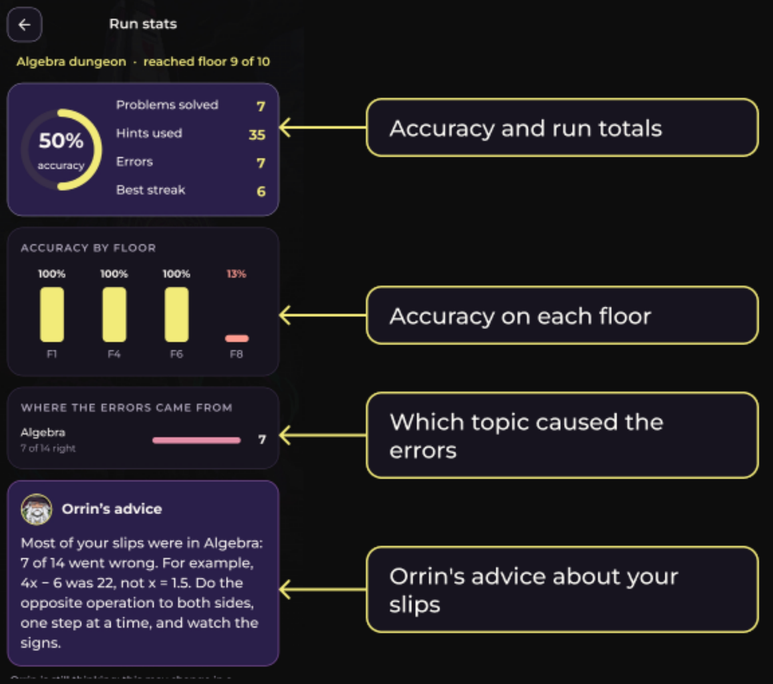
65. Boss fight (Boss battle screen)



66. Run end (Defeated screen)



67. Run end (Run stats screen)



Annotated Wireframes

Design Development

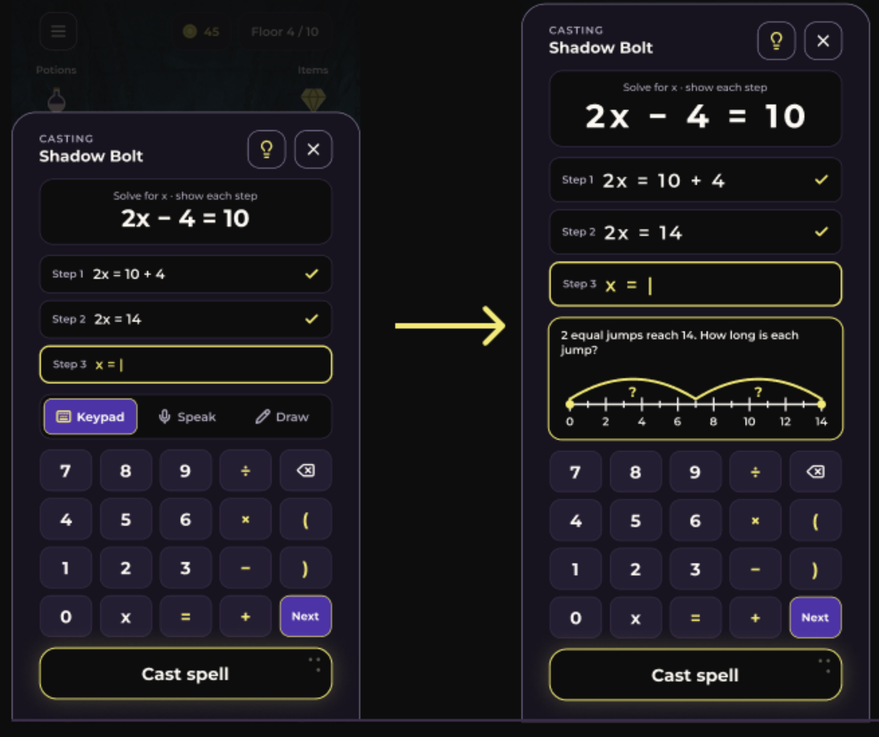
How the design grew from the first version, and what led to each change.

1 · Dyscalculia mode

Question in class

FIRST VERSION

NOW



FIRST VERSION

Every sum was shown as plain text, the same way for every learner.

WHAT CHANGED AND WHY

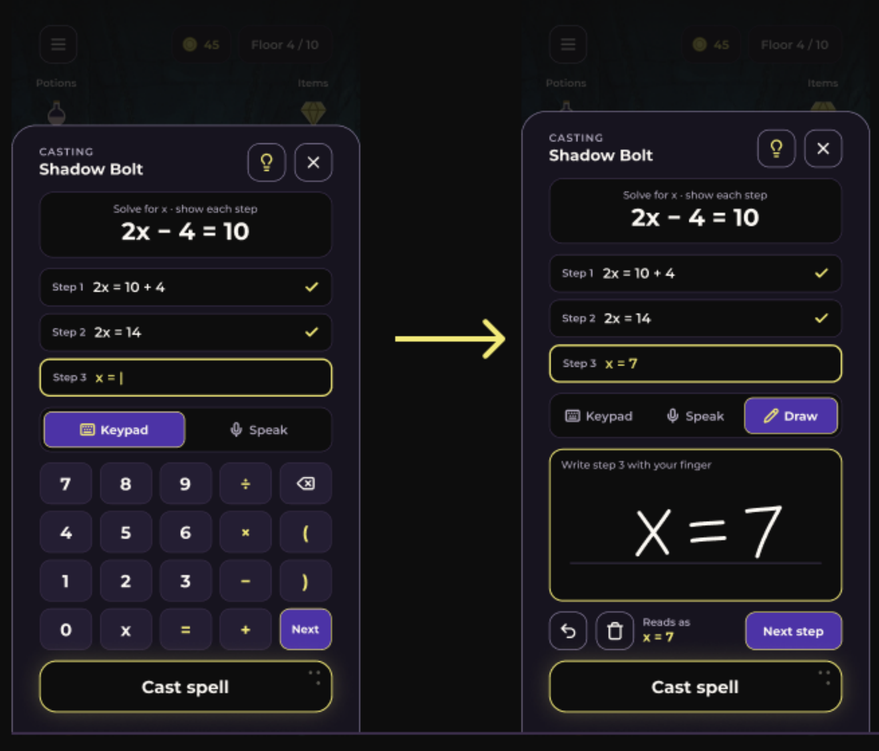
A student asked in class about learners with dyscalculia. After researching it, a Dyscalculia mode was added with bigger, spaced-out numbers and a picture of each sum.

2 · Ways to answer

Peer feedback

FIRST VERSION

NOW



FIRST VERSION

Players could only answer by typing on the keypad or by speech to text.

WHAT CHANGED AND WHY

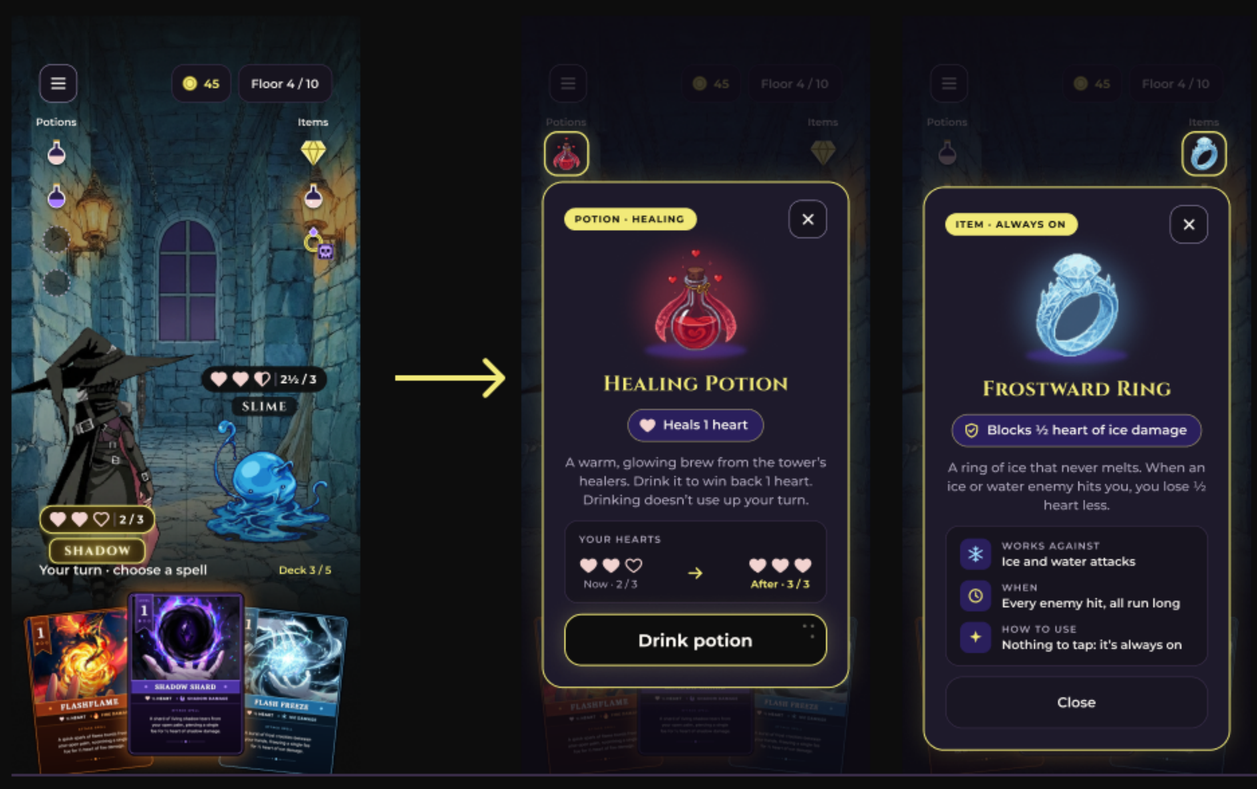
A fellow student pointed out that drawing can help learners who find typing maths hard, so a Draw option was added to write the step by hand.

3 · Potions and items

Design decision

FIRST VERSION

NOW



FIRST VERSION

Potions and items were only shown as icons at the side of the fight, with no way to see what they do.

WHAT CHANGED AND WHY

Tapping one now opens a pop-up that explains it, so the information is easy to find before a potion is used.

4 · Mixed maths

Research

FIRST VERSION

NOW

FIRST VERSION

Each run only trained one maths topic at a time, chosen before entering the tower.

WHAT CHANGED AND WHY

The research showed some students do not want to practise one topic at a time, so a Mixed dungeon shuffles every topic each floor.